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Laguna State Polytechnic University

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COLLEGE OF COMPUTER STUDIES

C# Console Application Final Project Proposal

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Project Title: NumSys Pro: Multi-Base Number System Calculator	
Introduction	The topic of manual conversion between binary, decimal, octal, and hexadecimal can be time-consuming and hard to understand. That is why this project proposes a C# Console Application that automates number system conversion, validates user input, displays accurate results, and shows the process. It aims to help students convert numbers quickly and to understand basic computing concepts. The program will be a useful learning tool for students, especially for those new to number systems and computer programming. It can save the time of manual calculation, reduce errors and make the conversion process easier to follow. This project helps the students to improve their problem solving skills, understand the number systems in a better way and get more confidence while using these ideas in their programming activities.
Statement of the Problem	Manual number conversion can cause errors and difficulties for programming students. This project addresses the following questions: 1. How can the application simplify number system conversion? 2. How can it validate inputs and produce accurate results? 3. How can it help students understand number conversion efficiently?
Objective of the Study	**A. General Objectives** ● To develop a C# Console Application that converts numbers between binary, decimal, octal, and hexadecimal. ● To improve the understanding of different number systems and their applications in computing. ● To create a simple and user-friendly program that provides fast and accurate number conversions. **B. Specific Objectives** ● To accept user input and number base selections. ● To convert numbers between the four number systems. ● To validate inputs and handle invalid entries. ● To apply basic C# programming concepts, including variables, conditional statements, loops, and methods. ● To display accurate results and test the application's functionality.

Scope and Limitation of the Study	**A. Scope** The proposed Number System Calculator will focus on converting whole numbers between binary (base 2), decimal (base 10), octal (base 8), and hexadecimal (base 16). It will allow users to enter numbers, select its original number system, choose a target number system, and view converted results. The system will accept valid whole-number inputs and convert them among the different number systems. The application will have an input validation to identify invalid digits based on the selected number system. It will also provide options for users to perform another conversion or exit the program. The system will use basic C# programming concepts, including variables, data types, console input and output, conditional statements, looping statements, and methods where appropriate. **B. Limitations** The application will be limited to a console-based interface and basic whole number conversions. It will not include a graphical user interface, web integration, mobile application development, complex database management, or online transactions. The supported range of numbers will depend on the data types and conversion methods used in the program. The application will serve as a learning aid and a tool for checking answers, but it will not replace the need for students to understand and practice manual number system conversion. The system will focus only on basic number conversion and will not include advanced mathematical operations or other features beyond the project intended functions.
Review Literatures and System	**Limitations of existing systems** The existing converters reviewed emphasize speed and convenience. For example, one app advertises accurate and instant results, an easy-to-use interface, and offline use (Younas, n.d.). The descriptions of these apps focus on what they output and do not mention showing the steps of a conversion. This suggests that explaining the process may not be their main purpose. This is further supported by an open-source converter whose developer lists an educational mode with step-by-step explanations as a feature still to be added (Maxcoder-svg, n.d.). C#'s built-in conversion method also returns only the final value, and it is limited to bases 2, 8, 10, and 16, with negative numbers interpreted using two's complement (Microsoft, n.d.). Lastly, the existing tools are packaged as finished mobile or web products. They are not presented as programming exercises, so they do not show students how conversion logic can be built with basic constructs such as loops, conditionals, and methods. NumSys Pro addresses these gaps by being a console application that shows the process and applies those constructs.
References:	1. Microsoft. (n.d.). Convert.ToString method (System). Microsoft Learn. https://learn.microsoft.com/en-us/dotnet/api/system.convert.tostring 2. Younas, M. A. (n.d.). Binary to Decimal Converter [Mobile application]. Apple App Store. https://apps.apple.com/jm/app/binary-to-decimal-converter/id6736915523 3. Maxcoder-svg. (n.d.). [Exact repository title still needs verification]. GitHub. https://github.com/Maxcoder-svg

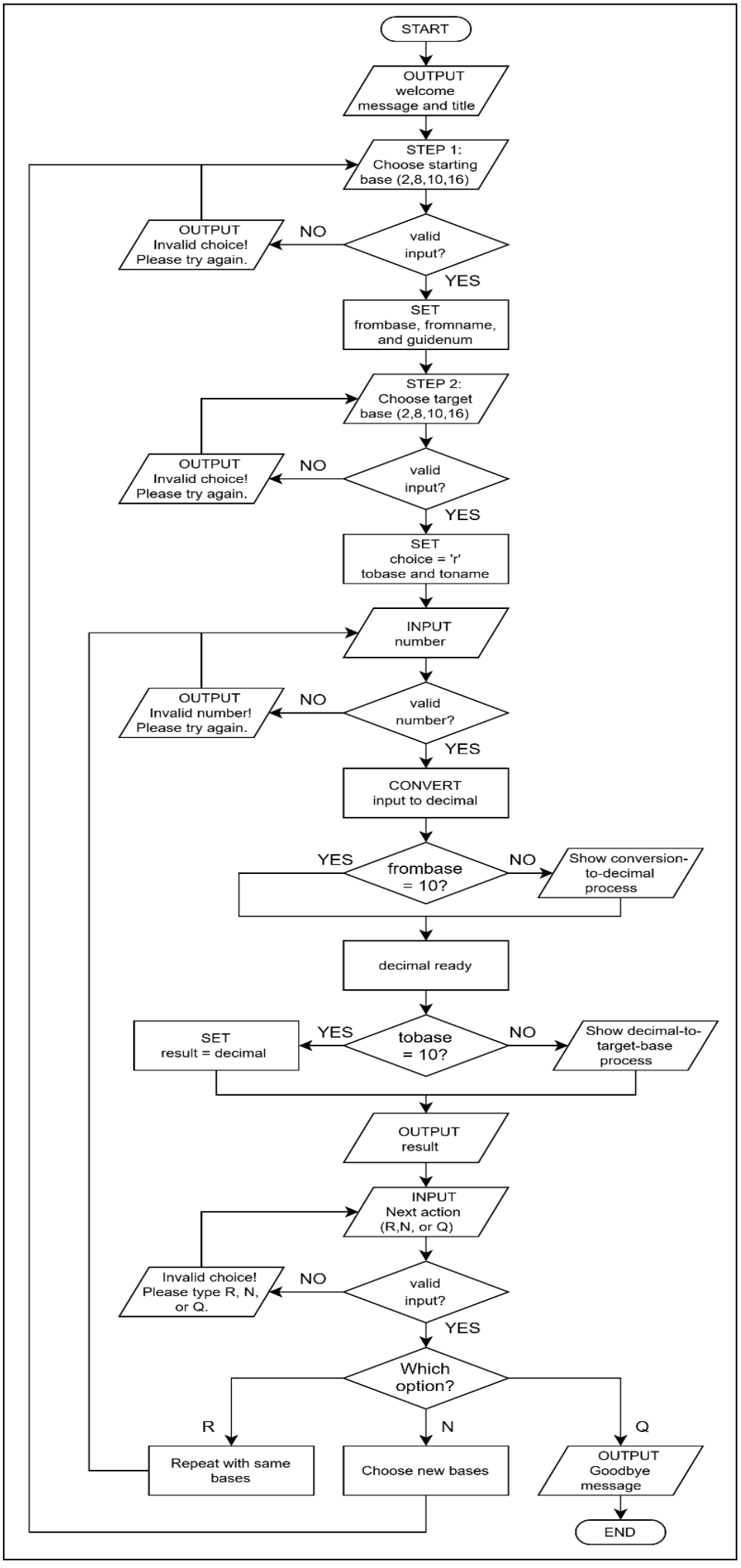
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Business and Functional Requirements

Business Requirement ID	**Business Requirement Description**	**Functional Requirement ID**	**Functional Requirement Description**
BR-001	Have a user-friendly interface.	FR-001	The system shall display the available number systems (binary, octal, decimal, hexadecimal) as numbered options before asking for input.
		FR-002	The system shall display the valid digits for the chosen number system before the user enters a number.
BR-002	Make the process of converting numbers from one number system to another easier.	FR-003	The system shall allow the user to choose the number system to convert from and the number system to convert to.
		FR-004	The system shall allow the user to enter a number to be converted.
		FR-005	The system shall show the mathematical process and computation of conversion.
		FR-007	The system shall have the options to retry, pick new bases, or quit.
BR-003	Ensure accurate and reliable number system conversions.	FR-008	The system shall correctly convert binary, decimal, octal, and hexadecimal numbers using their respective formulas and processes on how to do so.
		FR-009	The system shall validate the user's input and display an error message if the entered number is invalid.

Flowchart


START
OUTPUT welcome message and title
STEP 1: Choose starting base (2,8,10,16)
valid input?
NO
OUTPUT Invalid choice! Please try again.
YES
SET frombase, fromname, and guidenum
STEP 2: Choose target base (2,8,10,16)
valid input?
NO
OUTPUT Invalid choice! Please try again.
YES
SET choice = 'r' tobase and toname
INPUT number
valid number?
NO
OUTPUT Invalid number! Please try again.
YES
CONVERT input to decimal
frombase = 10?
YES
NO
Show conversion-to-decimal process
decimal ready
tobase = 10?
YES
SET result = decimal
NO
Show decimal-to-target-base process
OUTPUT result
INPUT Next action (R,N, or Q)
valid input?
NO
Invalid choice! Please type R, N, or Q.
YES
Which option?
R
N
Q
Repeat with same bases
Choose new bases
OUTPUT Goodbye message
END

Pseudocode

```
START
DISPLAY welcome message and title

REPEAT   // returns here if user chose N

  REPEAT
    DISPLAY "STEP 1: Choose starting base (2, 8, 10, 16)"
    INPUT inputbase
    IF inputbase is not 2, 8, 10, or 16 THEN
      DISPLAY "Invalid choice! Please try again."
    END IF
  UNTIL inputbase is 2, 8, 10, or 16
  SET frombase, fromname, and guidenum based on inputbase

  REPEAT
    DISPLAY "STEP 2: Choose target base (2, 8, 10, 16)"
    INPUT inputbase
    IF inputbase is not 2, 8, 10, or 16 THEN
      DISPLAY "Invalid choice! Please try again."
    END IF
  UNTIL inputbase is 2, 8, 10, or 16
  SET choice = "R"
  SET tobase and toname based on inputbase

  REPEAT   // returns here when the user chooses R

    // number inputting section
    REPEAT
      DISPLAY "Enter a" fromname "number"
      DISPLAY "Use only" guidenum
      INPUT number
      IF number is empty, has a digit not allowed in frombase, or is too big THEN
        DISPLAY "Invalid number! Please try again."
      END IF
    UNTIL number is valid

    // convert input to decimal section
    PROCESS number into decimal
    IF frombase is not 10 THEN
      DISPLAY conversion-to-decimal process (digit x base^position)
    END IF
    // decimal ready

    // Convert decimal to target base
    IF tobase = 10 THEN
      SET result = decimal
    ELSE
      PROCESS decimal into result by dividing by tobase and collecting the remainders
      DISPLAY decimal-to-target-base process
    END IF

    DISPLAY result

    // choosing next action
    REPEAT
      DISPLAY "R - Retry with same bases"
      DISPLAY "N - Choose new bases"
      DISPLAY "Q - Quit"
      INPUT choice
      IF choice is not R, N, or Q THEN
        DISPLAY "Invalid choice! Please type R, N, or Q."
      END IF
```

```
        UNTIL choice is R, N, or Q

    UNTIL choice is not "R"

UNTIL choice = "Q"

DISPLAY goodbye message
END
```